

C2T5

D.I.A.L SYSTEM

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Use Cases

General

1. Sign up
2. Login

DAS 1

- 1: Digital literacy screening tests
2. Dashboards summarising students' performance

DAS 3

1. Track child's progress
2. Notify parent

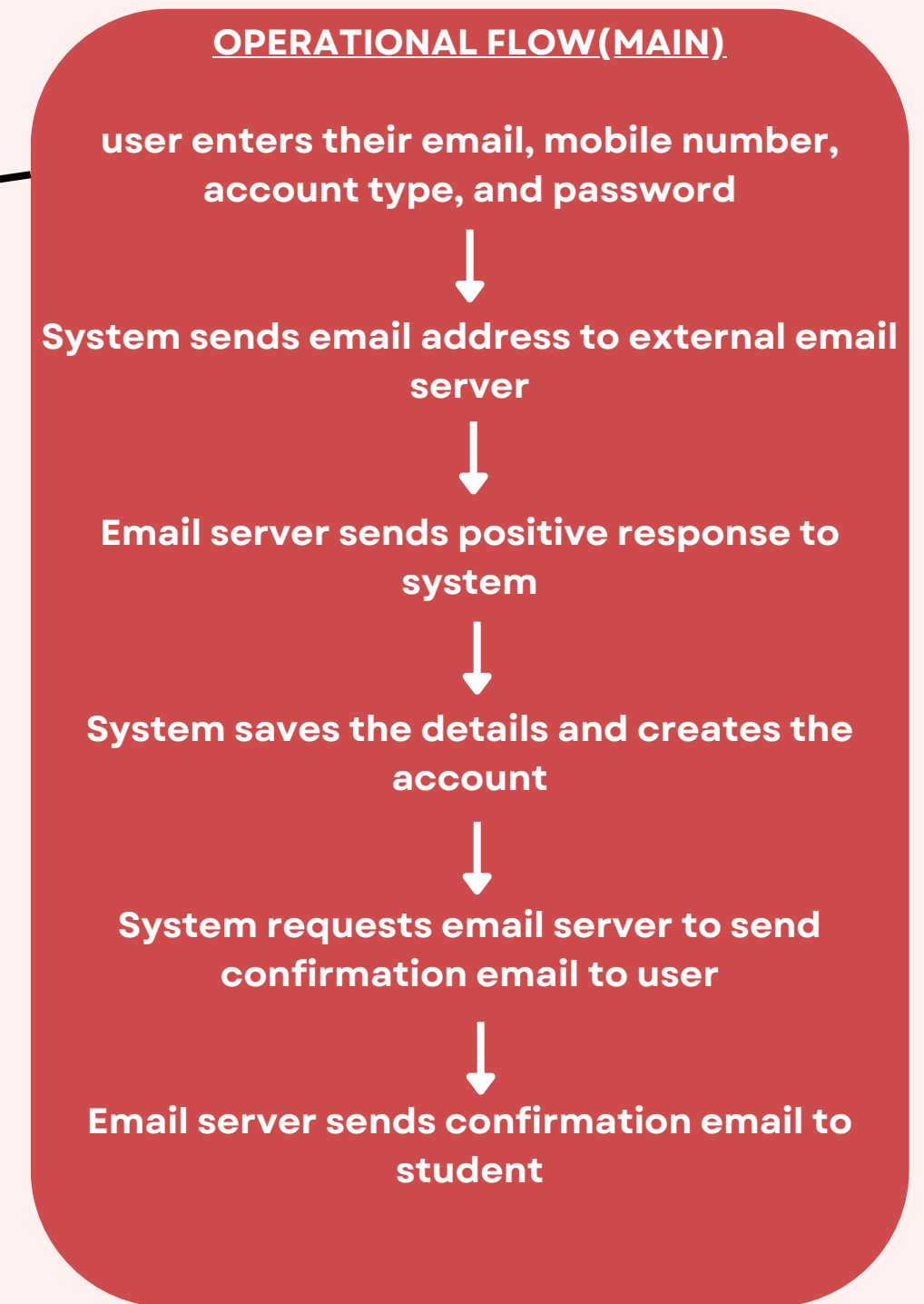
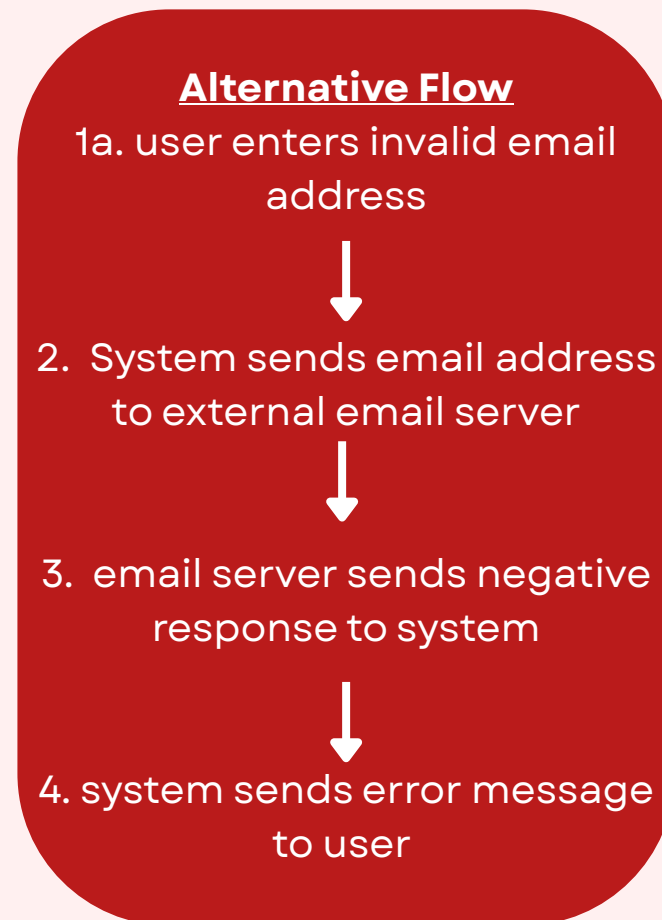
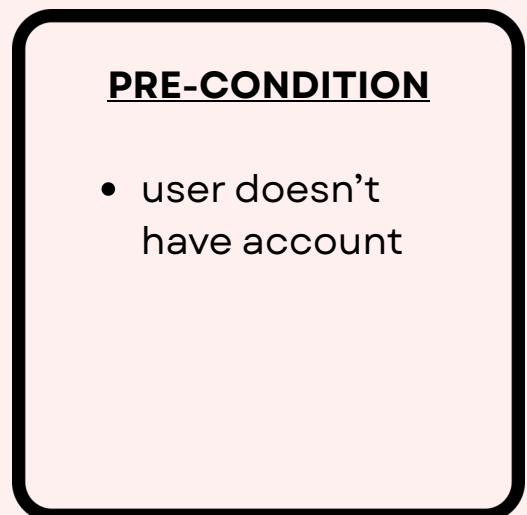
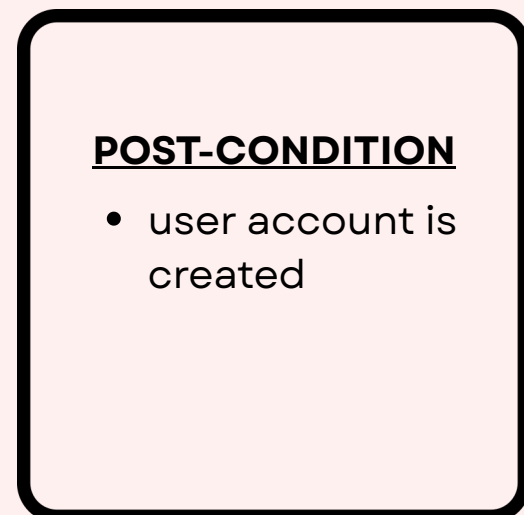
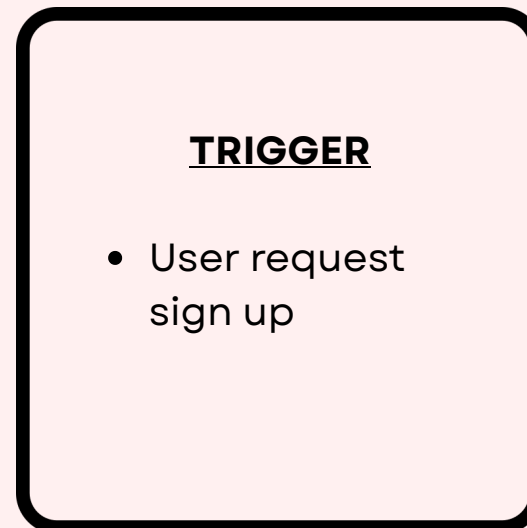
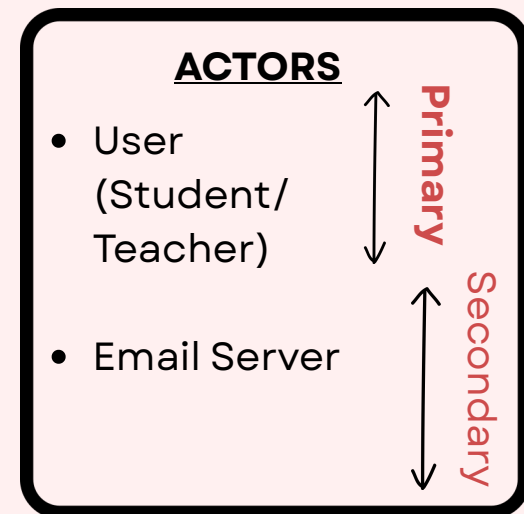
DAS 7

1. Generate worksheet for learning
2. Track child's educational progress
3. Email parents to check their child's progress



What are our Use Cases?

Use Cases 1: Sign Up



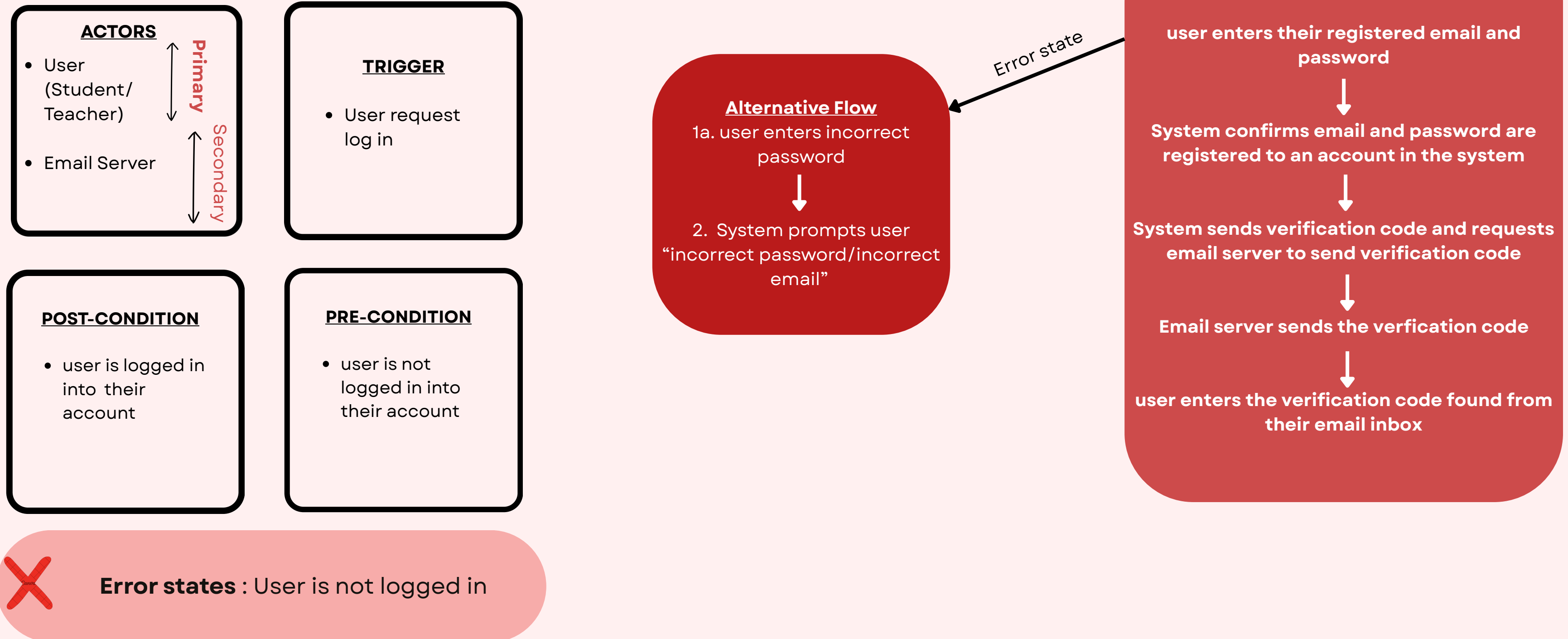
Error state



Error states : Account is not created

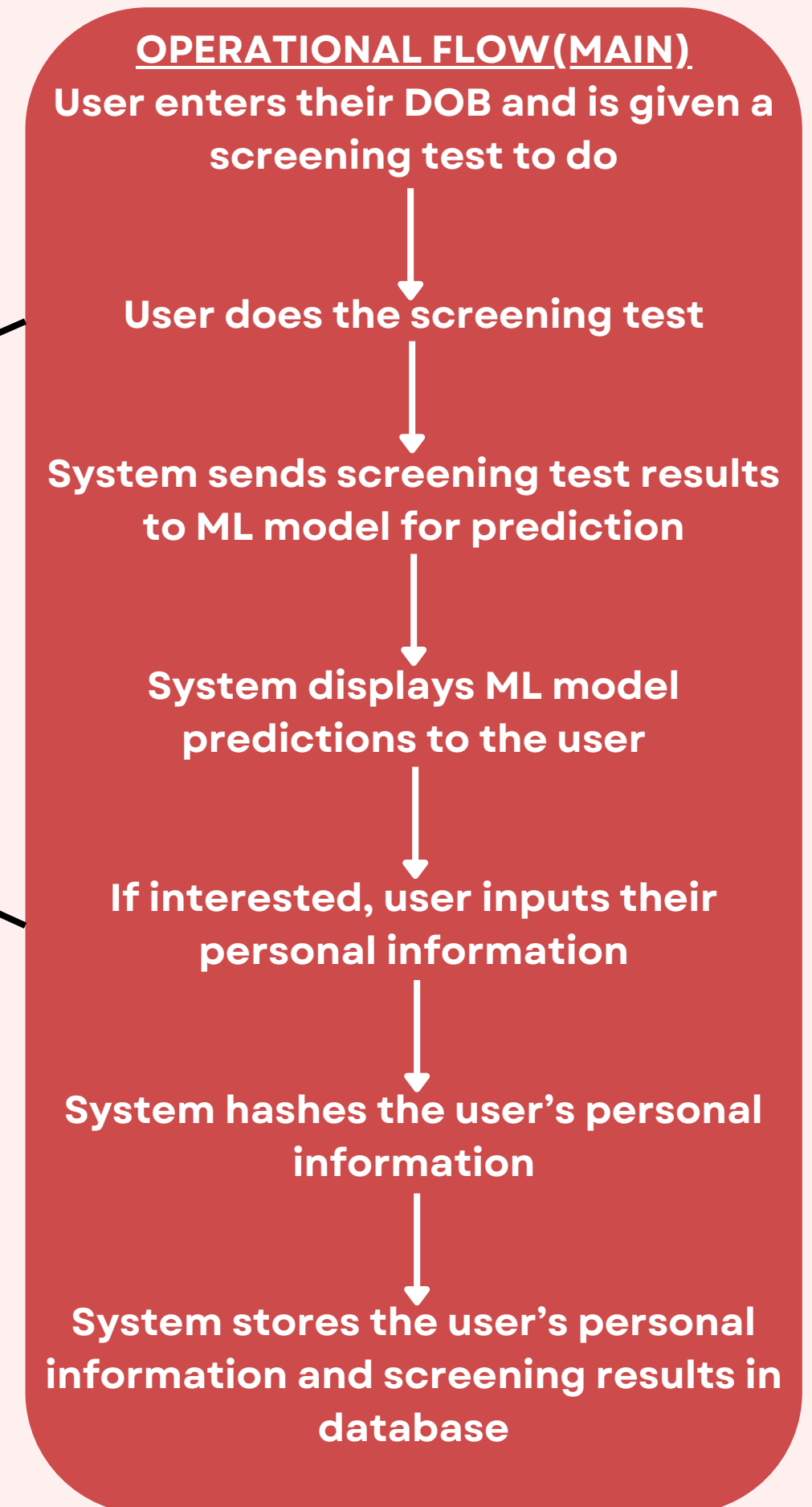
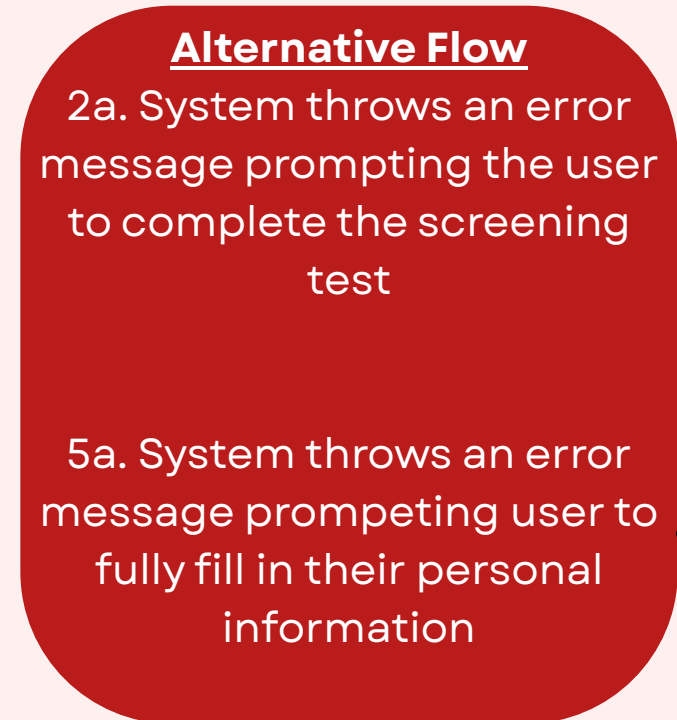
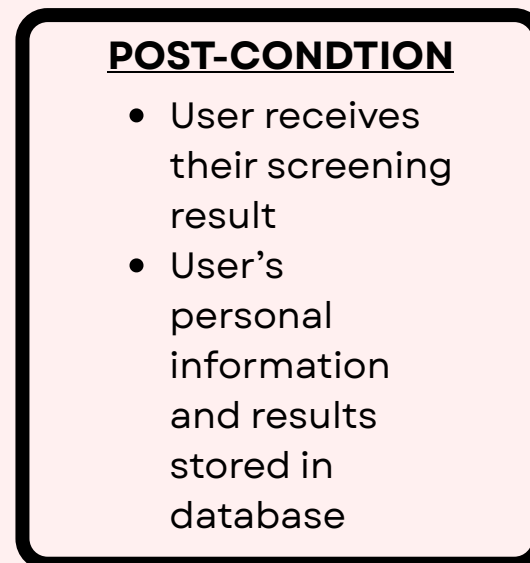
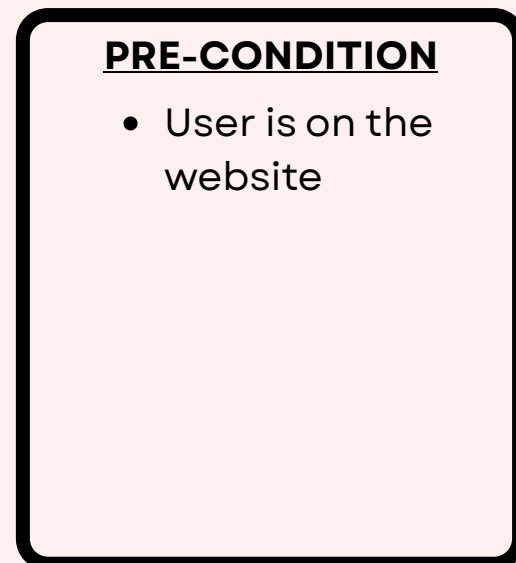
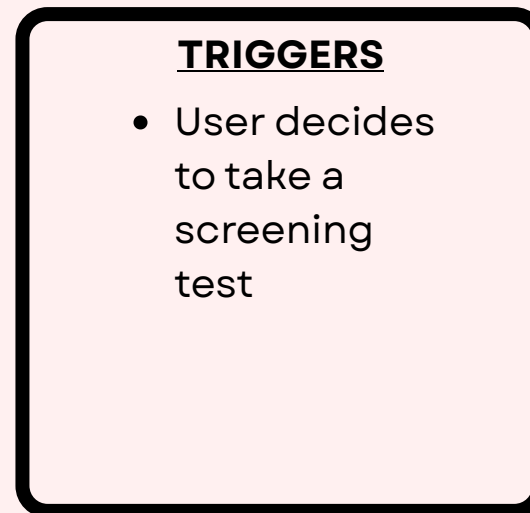
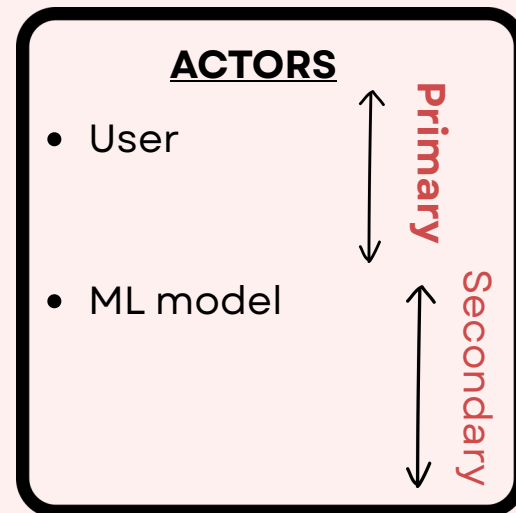
What are our Use Cases?

Use Cases 2: Log in



What are our Use Cases?

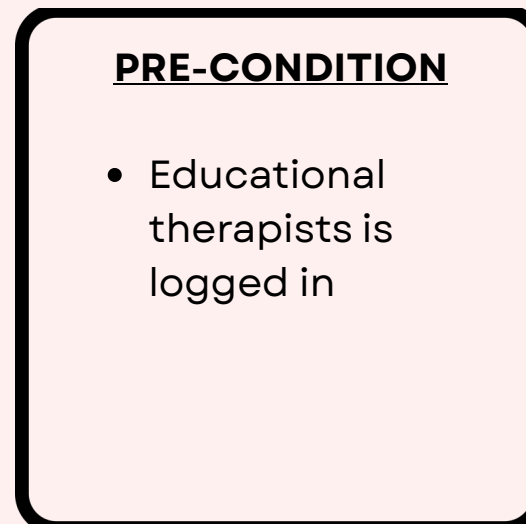
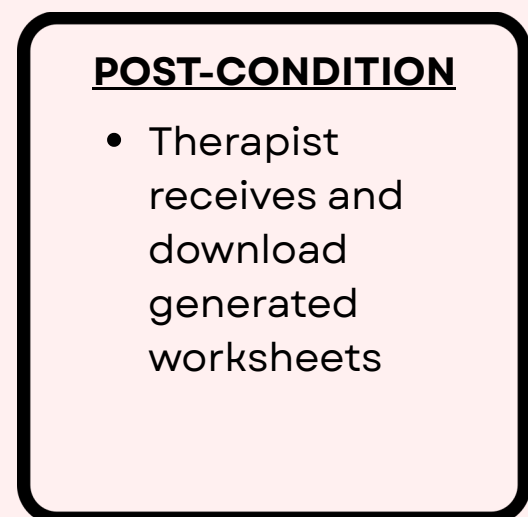
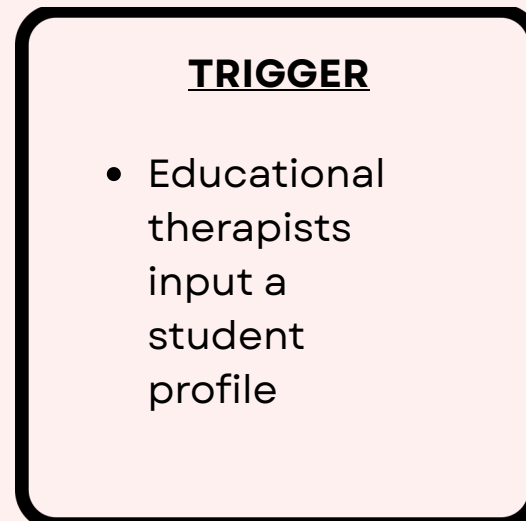
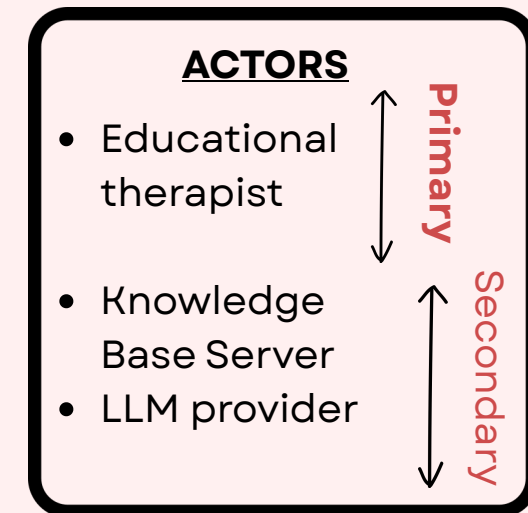
Use Case: Screen User



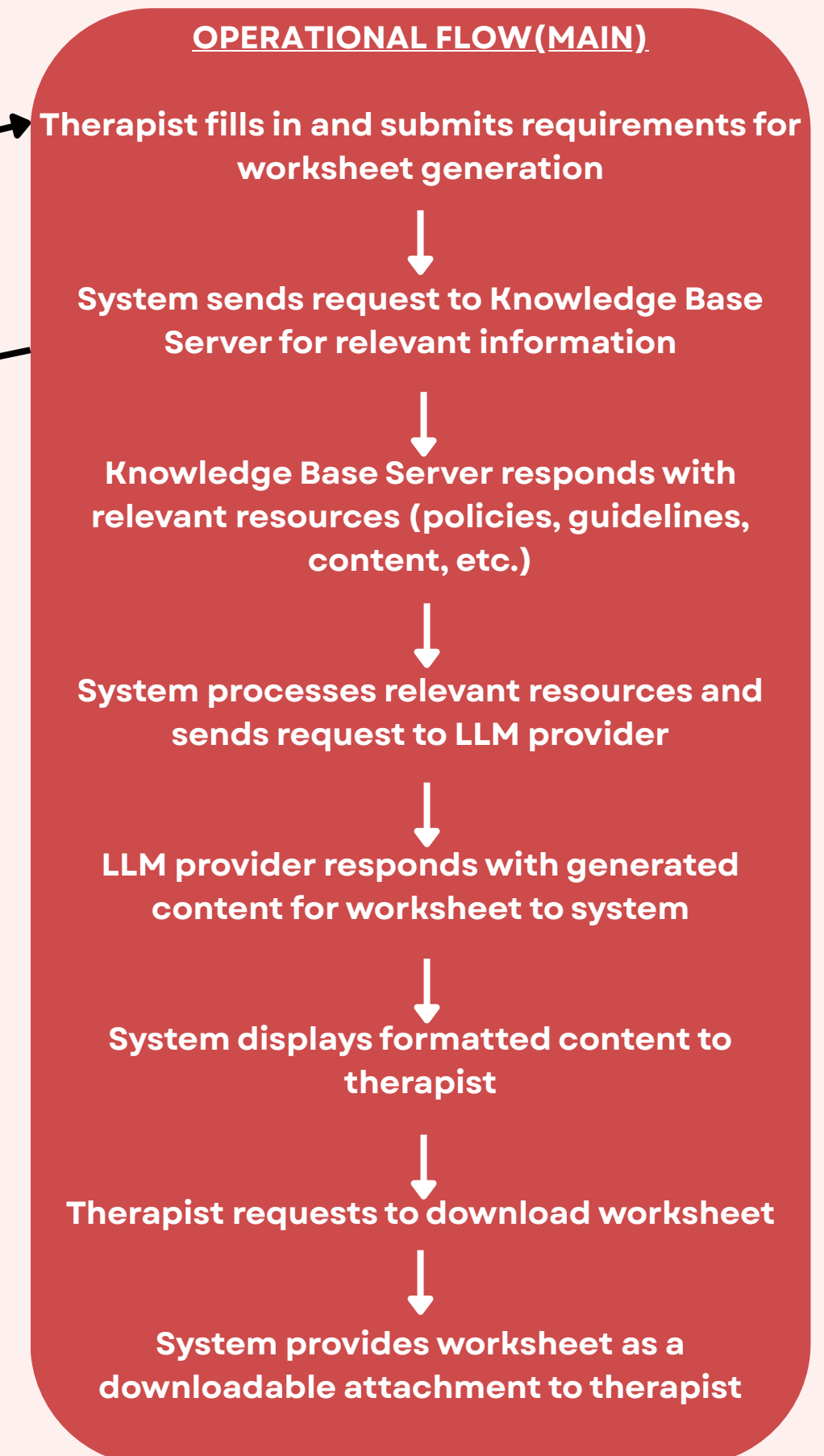
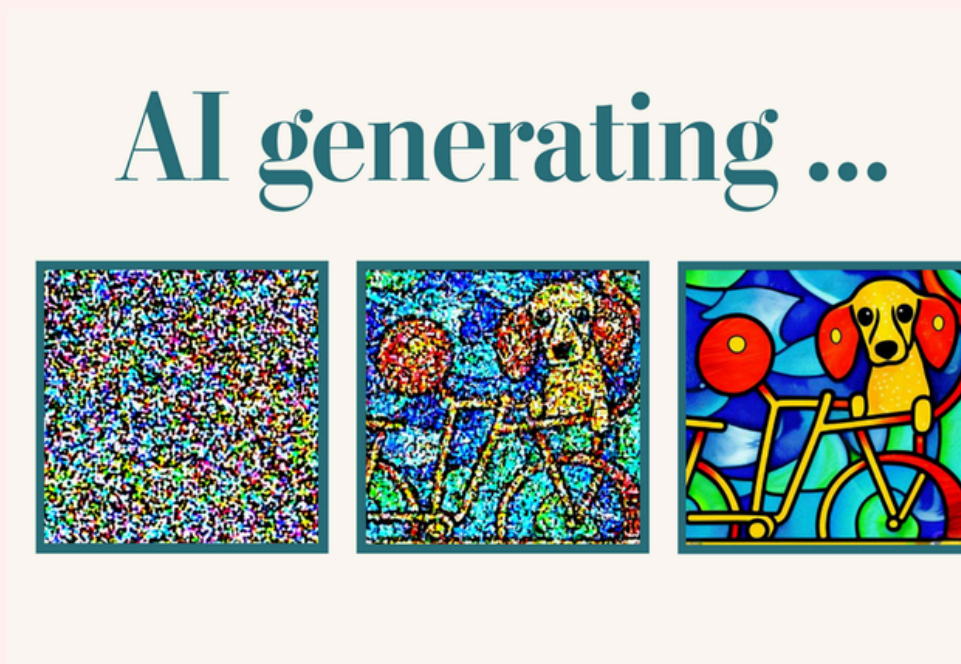
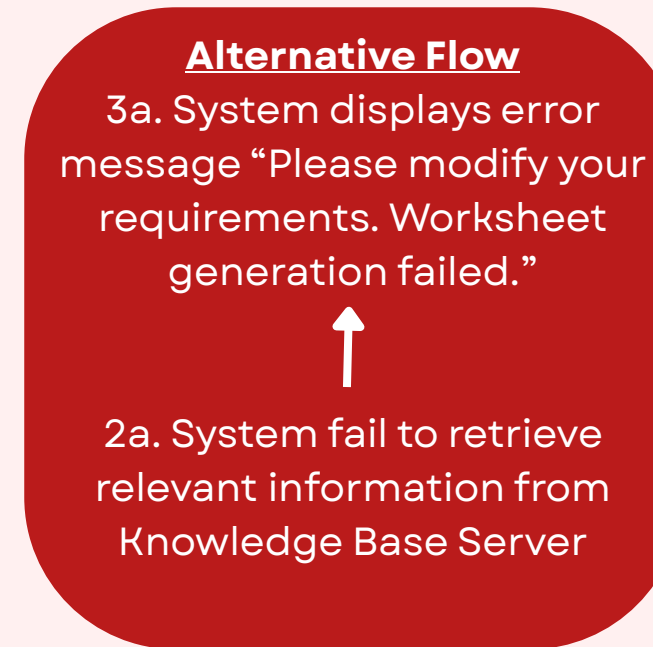
Error states : User fails to fully complete screening test / filling in their personal information before submitting

What are our Use Cases?

Use Cases 3: generate worksheet

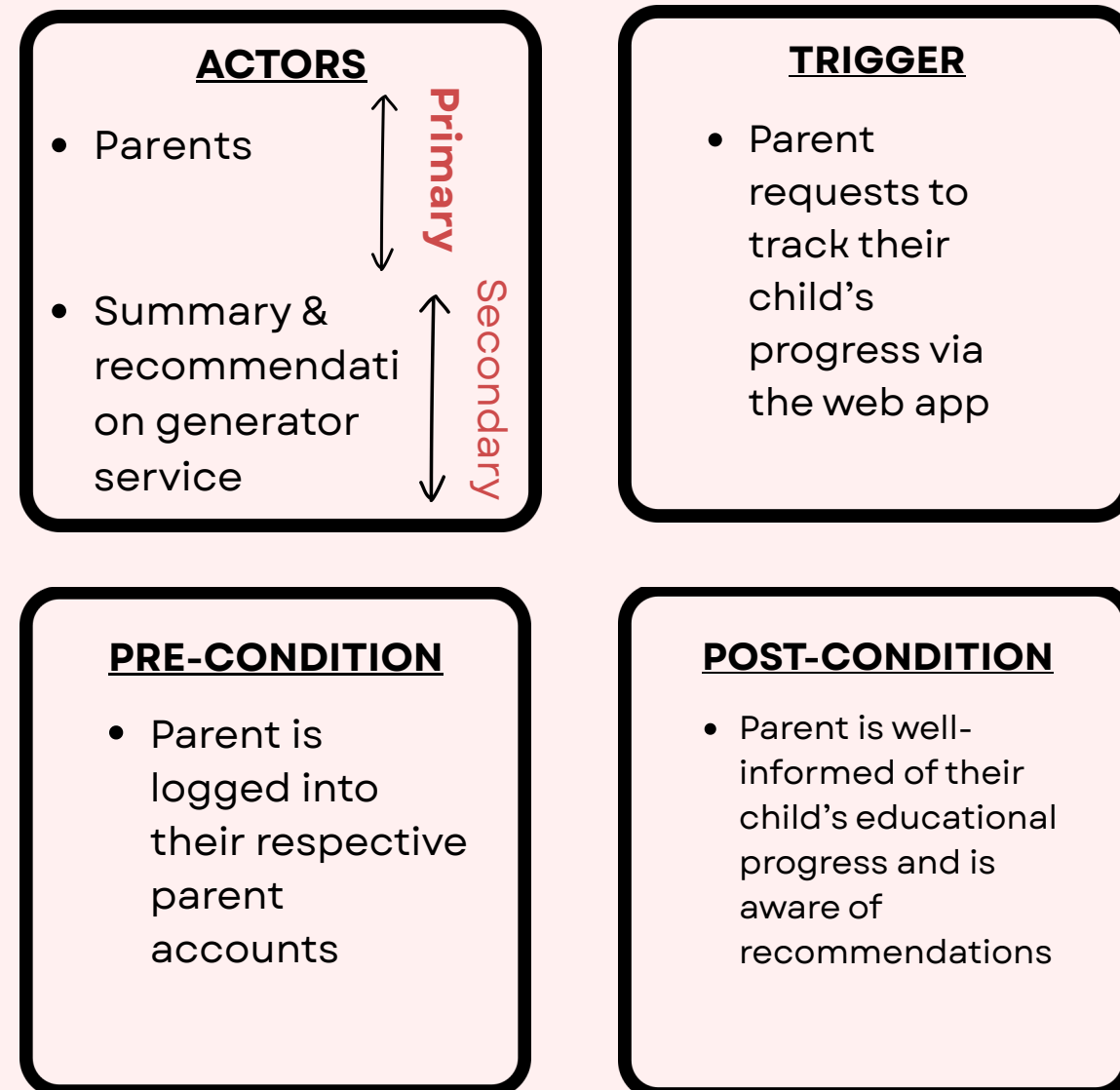


✗ **Error states** : Worksheet generation failed

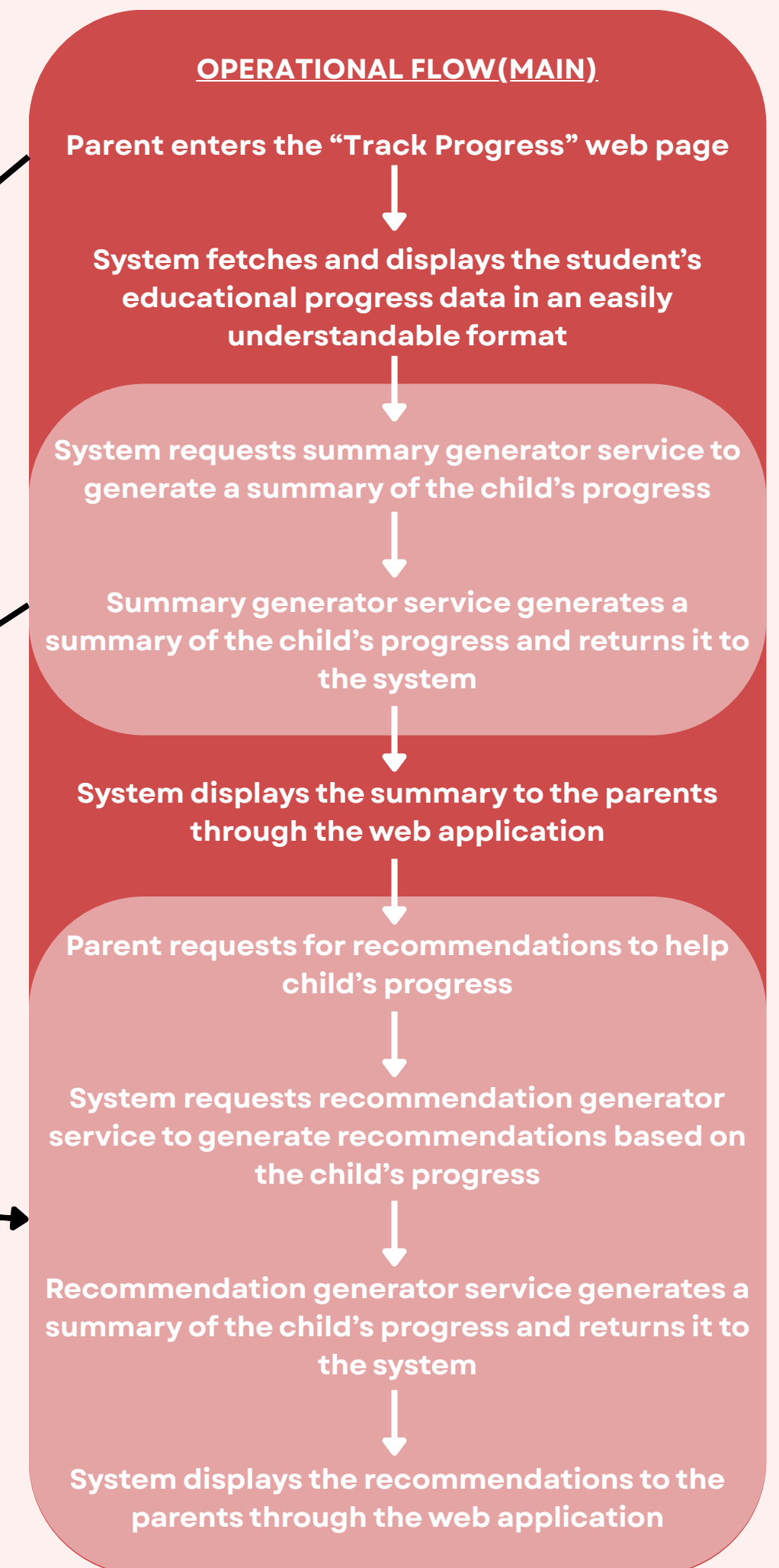
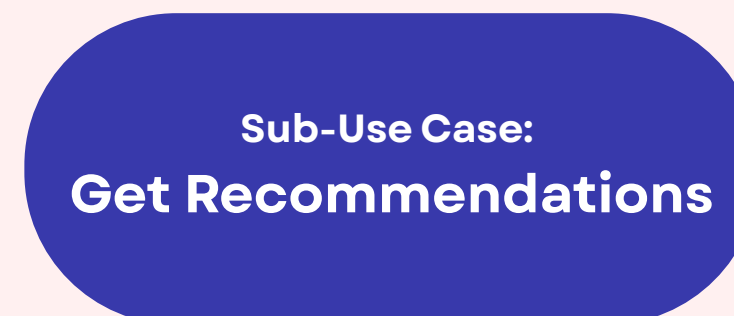
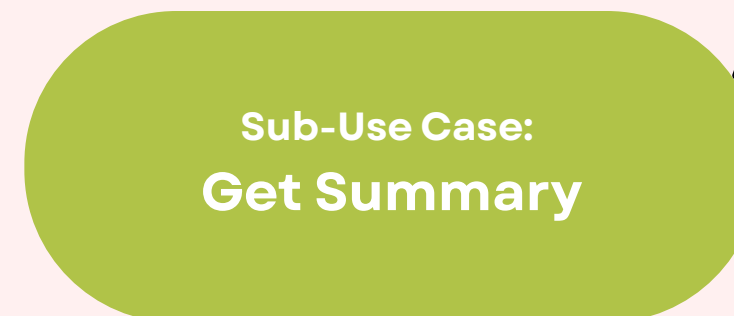
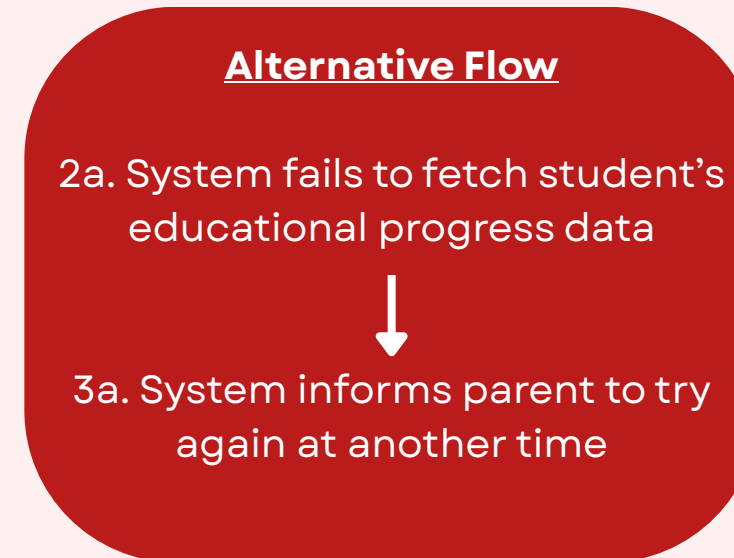


What are our Use Cases?

Use Cases DAS 7: Track child's progress

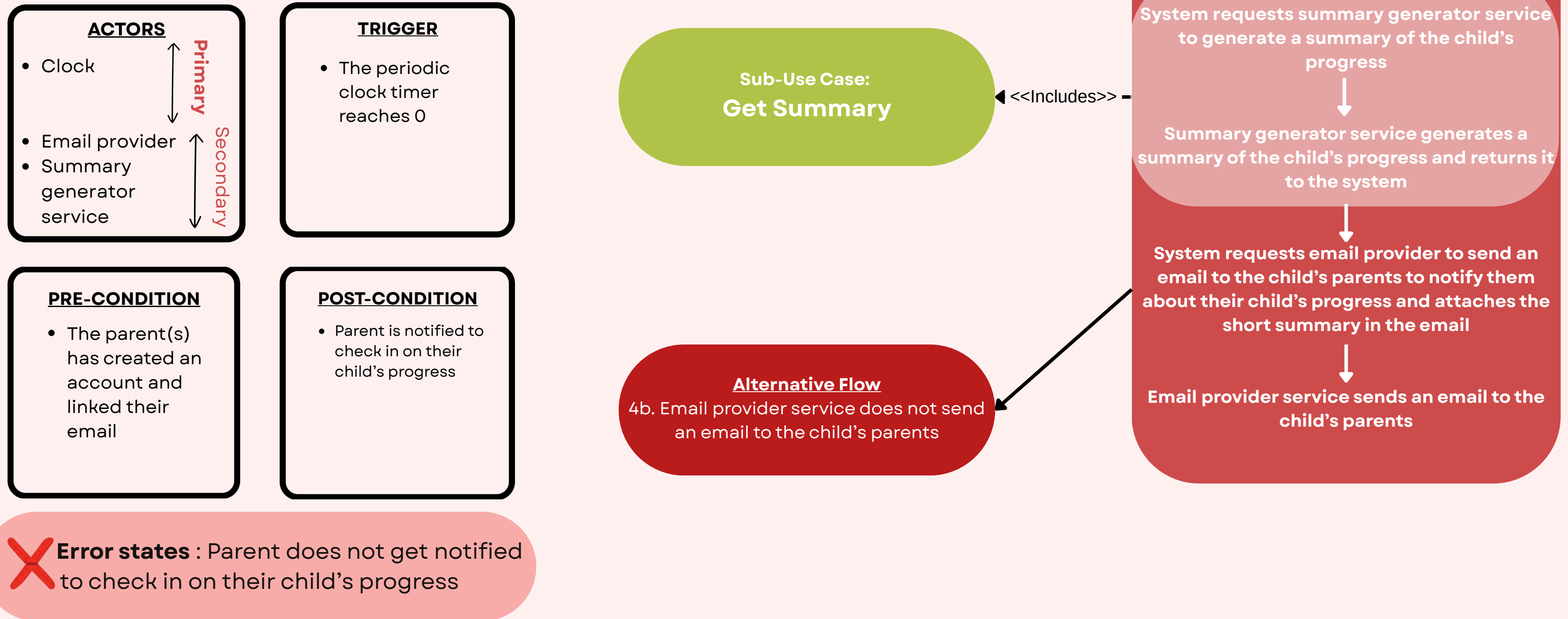


Error states : Parent is unable to access their child's progress



What are our Use Cases?

Use Cases DAS 7: Notify Parent



Timeline

Week 7

- Finalize features and flows (Generate related content activities,)
- Initialize Git repos for frontend and backend
- Set up project
- Implement basic REST APIs
- Web UI layouts: basic dashboard, basic chat interface
- AI architecture and prompt pipeline
- RAG workflow and retrieval strategy

Week 8 (PM2 deliverables ready)

- Connect UI → backend via JSON over REST
- Backend: refine API responses with correct success and error messages
- Implement prompt button
- Begin developing evaluation framework for
- Blackbox Unit testing backend and frontend

Week 9 – 10 (PM3 deliverables ready)

- Integration testing
- Verify RAG delivery accuracy
- Develop tests for error states
- System end-to-end testing

Week 11-12

- Robustness testing
- Develop test to bypass prompt guidelines and check for irrelevant content generation
- Work on report and documentation

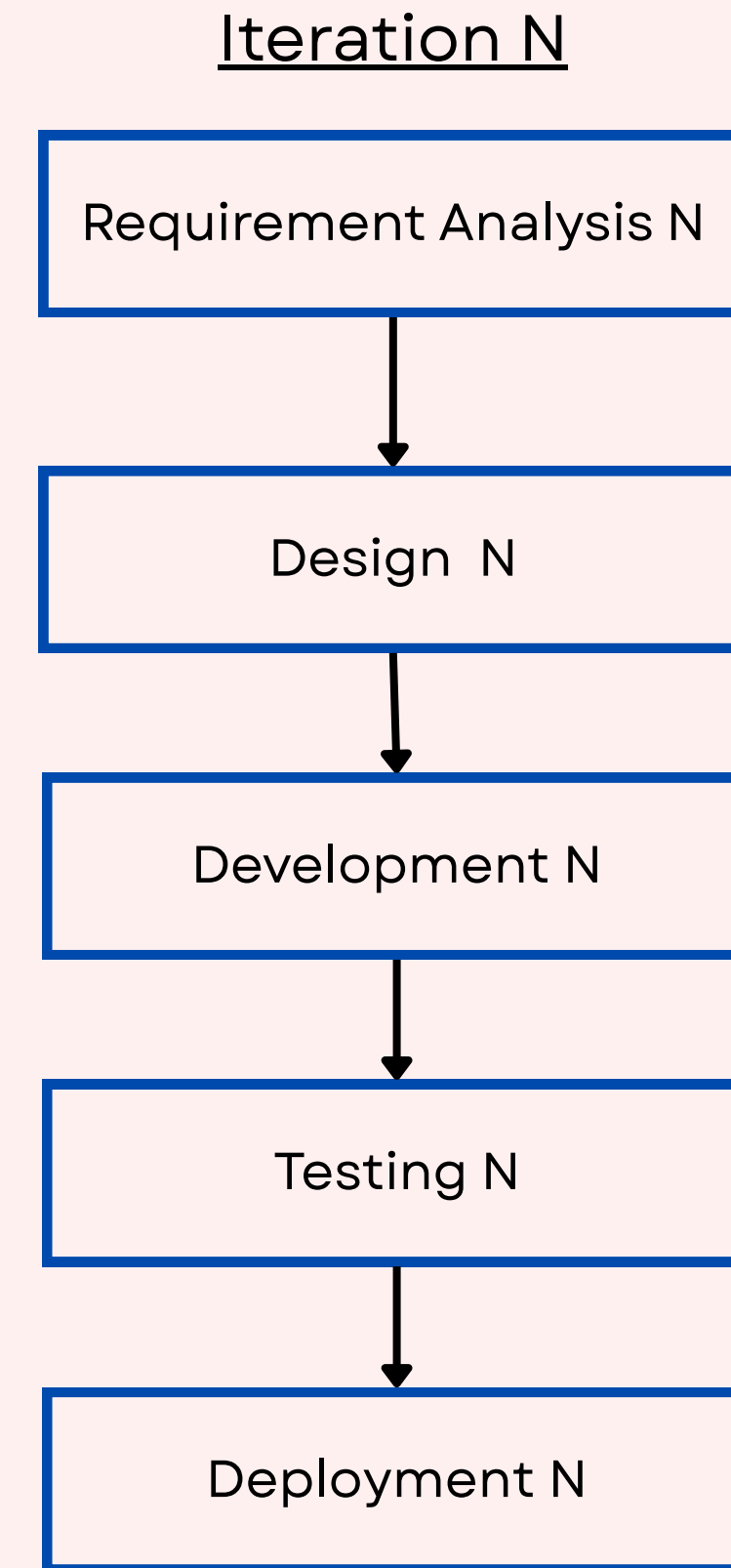


Software Development Process

- **Iterative model**

Justification:

- Both projects involve AI/ML components (summarisation, literacy risk prediction) whose behaviour and accuracy are hard to predict upfront thus a fixed one-shot plan is unrealistic.
- User testing is explicitly listed as a deliverable, implying feedback loops are expected. This means the team gets real feedback on whether the output is actually useful and understandable to non-technical users which cannot be validated through planning alone. The project adapts based on this feedback in each subsequent iteration.
- Since these are student projects with a hard submission deadline, the Iterative Model's time-boxing ensures that if a feature takes longer than expected, it is deferred to the next iteration rather than delaying the entire project. This keeps the team on track and ensures a working product is always available at the end of each cycle, even if not all features are complete.
- These are student projects with a fixed scope and deadline hence the frequent requirement change assumption of Agile methods like Scrum/XP is unsuitable.



$N \in \mathbb{N}^+$



Software Development Process (iterative)

- **DAS 1**

Iteration N breakdown:

- (1) User screening interface
 - Build student UI to assign tasks and capture responses
- (2) Database and ML classification model
 - Store collected response and train ML model
 - Predict literacy risk indicators
- (3) Educator dashboard
 - generate readable summaries

Potential risks:

- (1) Poor/ unavailable datasets
- (2) Inadequate ML/AI knowledge
- (3) Scope creep

Risk mitigation:

- (1) Clarify data access with DAS before Iteration 2. Use public literacy or synthetic datasets as backup
- (2) Use iteration 1 as reference. Define evaluation metrics early. Use pre-trained APIs.
- (3) Time-box iterations. Maintain a feature backlog. Define MVP for the demo early.



Software Development Process (iterative)

- **DAS 3**

Iteration N breakdown:

(1) Prepare

- Splitting documents into token chunks
- embed each chunk using an embedding matrix
- Store them into Vector database

(2) Retrival

- writing query to retrieve information from database

(3) Generation

- Prompt engineering and evaluation

Potential risks:

(1) Poor Chunking Strategy

(2) Irrelevant or Inaccurate Retrieval

(3) Prompt Hallucination Despite Retrieval

Risk mitigation:

(1) Experiment with different chunking strategies early fixed size, sentence based, or semantic chunking and evaluate which suits best for DAS.

(2) Evaluate retrieval precision separately from generation quality using test queries with known expected results.

(3) Engineer prompts that explicitly instruct the model to ground its response only in the provided context.



Software Development Process (iterative)

- **DAS 7**

Iteration N breakdown:

- (1) Dashboard UI
 - Create dashboard for parents
 - Establish role-based security access
- (2) Data visualisation and Database integration
 - Connect live student progress data
- (3) AI summarisation
 - Integrate LLM api

Potential risks:

- (1) Data privacy
- (2) Inadequate AI knowledge
- (3) Scope creep

Risk mitigation:

- (1) Implement authentication in iteration 1. Use firebase auth or JSON web token. Anonymise student data during development.
- (2) Use LLM APIs over custom NLP.
- (3) Time-box iterations. Define MVP.

Correspondence with Client

Clarifications Regarding the DIAL Project (SUTD 50.003 ESC)

Summarize this email

S

Student - Brian Wong Wei Xiang

To: soofrina@das.org.sg <Soofrina@das.org.sg>

Cc: Student - Vincent Alexander Yauvira; Student - Toh Shijie (Zhuo Shijie); Student - Soh Zheng Hean, Michael (Su Jingheng); Student - Lee Le Bin; +5 others

Wed 6/17/2026 8:00 AM

Dear Ms. Soofrina Binte Mubarak,

We are writing this email to clarify some points regarding the “DIAL” project collaboration between SUTD 50.003 ESC and DAS.

As we are mapping out the technical and functional requirements for the system, we wanted to ask a few questions to ensure our development aligns with your expectations and data privacy standards. To make this easier to review, we have grouped our inquiries by their respective sub-systems.

Sub-system 1: DAS Learning Screening Engine

Training Data & Sharing: Will datasets be provided to train the ML/AI models for this specific sub-system along side the guidelines on how to correctly interpret the results? If so, are we permitted to upload this data to online LLMs or public file-sharing platforms like GitHub, or are there strict data protection protocols we must follow?

Data Anonymization: When storing screening results, what are the exact requirements for anonymization? Since we still need to associate results with a user to process referrals, would using a hashed user ID be considered sufficiently anonymized?

Access Control: Is this screening test meant to be freely accessible to anyone who creates an account, or will it only be assigned by teachers to specific students?

Sub-system 3: DAS Adaptive Learning Activity Generator

Training Data & RAG: Similar to sub-system 1, will specific datasets be provided for training this generator? Additionally, will a curated database be provided for us to use with the Retrieval-Augmented Generation (RAG)?

Worksheet Generation: Will we be provided with appropriate instructional strategies, pedagogical guidelines, or specific learning objectives to shape how these adaptive worksheets are generated?

General Technical & Implementation Requirements

Hosting Environment & LLM Integration: Are we expected to host the LLMs and the website locally (particularly due to security and prompt injection concerns), or will DAS/SUTD provide a dedicated hosting service for us to use?

Performance Metrics: Are there specific target metrics (e.g., accuracy and recall thresholds) that we should be aiming for with the ML models across the project?

Thank you for your time and guidance. We look forward to your feedback so we can move forward confidently with the project.

Best regards,

Brian Wong

C2T5

Clarifications Regarding the DIAL Project (SUTD 50.003 ESC)

Summarize this email

SM

Soofrina Binte Mubarak<Soofrina@das.org.sg>

To: Student - Brian Wong Wei Xiang

Cc: Student - Vincent Alexander Yauvira; Student - Toh Shijie (Zhuo Shijie); Student - Soh Zheng Hean, Michael (Su Jingheng); Student - Lee Le Bin; +5 others

Thu 6/18/2026 10:10 AM

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Hi Brian,

Please see my responses to your questions: <https://docs.google.com/document/d/1CAsYajEaKEfLkPjg0ltqNYBoI-A-iJodAHlclkoAAuA/edit?usp=sharing>
We can use this document to continue Q&A if there are more.
There's one outstanding response - I have marked it in red because I need to check with my ICT colleagues. As this week is still a leave period, most colleagues are away from work but I should be able to get a response by next week.

Thank you!

Soofrina | Lead Educational Therapist | Assistant Director (EduTech) | RETA Fellow
English Language and Literacy (ELL) Division
Dyslexia Association of Singapore
73 Bukit Timah Road, #05-01, Singapore 229832
Tel: 65-6643 9600 | Fax: 65-643 9603
Dyslexia Association of Singapore | DAS International | DAS Academy
www.das.org.sg | www.dasint.org.sg | www.dasacademy.edu.sg

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Correspondence with Client

Summary

Sub-system 1: DAS Learning Screening Engine

Training Data & Sharing

Question: Will datasets be provided to train the ML/AI models, and can we upload this data to online LLMs or public platforms like GitHub?

Response: No existing data is available; the screener is a new, non-diagnostic "fun questionnaire" designed to indicate a likelihood of conditions like dyslexia or ADHD.

Data Anonymization

Question: What are the exact requirements for anonymization, and is a hashed user ID sufficient?

Response: Yes, a hashed ID is acceptable as long as individuals cannot be identified by their full name or NRIC.

Access Control

Question: Is this screening test freely accessible to anyone, or assigned by teachers?

Response: It will be freely accessible 24/7, routing users by age. The AI will analyze and store the results until the user proceeds with formal profiling.

Correspondence with Client

Summary

Sub-system 3: DAS Adaptive Learning Activity Generator

Training Data & RAG

Question: Will specific datasets be provided for training this generator, and will a curated database be provided for RAG?

Response: Materials (Docs, PDFs, Slides) are in Google Drive and tracked via a spreadsheet. Full access will be granted once an NDA is signed.

Worksheet Generation

Question: Will we be provided with appropriate instructional strategies, pedagogical guidelines, or specific learning objectives?

Response: Yes, a dedicated team member will assist with this next week. To manage the scope, you can focus on just one band level (e.g., Band A) to build a working proof of concept.

Correspondence with Client

Summary

General Technical & Implementation Requirements

Hosting Environment & LLM Integration

Question: Are we expected to host the LLMs and the website locally, or will DAS/SUTD provide a dedicated hosting service?

Response: Pending; the client is checking with their ICT colleagues before confirming.

Performance Metrics

Question: Are there specific target metrics (e.g., accuracy and recall thresholds) we should aim for?

Response: No strict metrics are required. An accuracy threshold of 75-80% is acceptable for the proof of concept.